

We would like to extend our heartfelt thanks to each of you for your time and effort in reviewing the document. Your input has been invaluable in refining and enhancing the quality of the work. For your convenience, all changes made to the document are highlighted in **red color**. Below, you will find our point-by-point responses to each of your comments.

Dear Dr. Cristian Alfonso Jiménez Castaño

We sincerely thank you for all the valuable comments and recommendations you have provided. Your observations have been fundamental for improving the quality and clarity of the work.

Point 1. *Document cohesion from the problem statement onwards.*

Response. The document has been restructured to improve cohesion between sections, especially in the methodology and presentation of results.

Point 2. *Lack of bibliographic support in several subsubsections of the state of the art.*

Response. Appropriate bibliographic references have been added in all subsections that lacked support.

Point 3. *Absence of justification for the use of convolutional networks vs. other architectures.*

Response. The methodological section has been expanded to include detailed justification for the choice of U-Net architecture and comparison with alternatives.

Point 4. *Results and conclusions presented in separate and disconnected chapters.*

Response. The results sections have been unified and the connection with the research objectives has been improved.

Point 5. *Figure 1-1: font size not recognizable at 100%.*

Response. The font size has been increased and the figure has been resized for better visibility.

Point 6. *Figure 1-2: small letters and page organization.*

Response. The figure has been resized and the layout has been reorganized to avoid blank spaces.

Point 7. *Lines 429 and 440: usability frequency percentage not clear.*

Response. The methodology and study universe have been clarified for the calculation of these percentages.

Point 8. *Figure 1-3: font size and accommodation.*

Response. The font size has been increased and the distribution on the page has been improved.

Point 9. *Excessive concentration on multiple annotators without considering other problems.*

Response. The section has been expanded to include a broader view of the state-of-the-art problems.

Point 10. *Research question without clear methodology to justify U-Net.*

Response. The methodological justification for the choice of U-Net has been developed.

Point 11. *Quick dismissal of other image types without justification.*

Response. Detailed justification has been added for the focus on histological images.

Point 12. *General objective does not begin with verb in infinitive.*

Response. The general objective has been reformulated to begin with a verb in infinitive.

Point 13. *Line 791: interpretability not reflected in specific objectives.*

Response. The specific objectives have been reformulated to explicitly include interpretability.

Point 14. *Line 792: very strong comparison that requires extensive experimental framework.*

Response. The scope of the comparison has been adjusted to be realistic with the available experimental framework.

Point 15. *Section advances too much without presenting methodologies.*

Response. This section has been moved to the end of the document, after the conclusions.

Point 16. *Figure 2-2: improve resolution and size.*

Response. The resolution has been improved and the figure has been resized.

Point 17. *Figure 2-3: remove black part of the interface.*

Response. The image has been cleaned to show only the pathological image.

Point 18. *Section 2.2.2: too empty.*

Response. It has been completed with visual examples and detailed explanations.

Point 19. *Section 2.3.1: inconsistency between labels and icons.*

Response. Inconsistencies have been corrected and the connection between paradigms has been improved.

Point 20. *Line 970: clarify that they are machine learning paradigms in general.*

Response. The scope of the mentioned paradigms has been clarified.

Point 21. *Subsection 2.3: too brief.*

Response. It has been expanded with additional development and bibliographic references.

Point 22. *Subsections 3.1.2 and 3.1.3: without bibliographic support.*

Response. Appropriate bibliographic references have been added.

Point 23. *Incorrect order of subsections.*

Response. It has been reordered to follow the correct logical sequence.

Point 24. *Subsection 3.3: contribution not clear.*

Response. The purpose and contribution of this section have been clarified.

Point 25. *Table 3-1: organize for better visibility.*

Response. The table has been reorganized to improve readability.

Point 26. *Equation 4-10 too long.*

Response. It has been divided into intermediate functions for better readability.

Point 27. *Section 4.1.4: should not be there without experiments.*

Response. It has been merged with the previous section for better coherence.

Point 28. *Numbering begins at zero.*

Response. The numbering has been corrected to begin at one.

Point 29. *Figures 5-1 and 5-2 seem identical.*

Response. The duplicate figures have been unified.

Point 30. *Lines 1627-1629: it is not clear if the regularization terms are scalars.*

Response. The nature of the regularization terms in equation 5-1 has been clarified.

Dear Dr. Juan Bernardo Gómez Mendoza

We sincerely thank you for your positive evaluation and all the valuable comments and recommendations you have provided. Your observations have been crucial for improving the quality and completeness of the work.

Point 1. *Lack of specific tests to illustrate the effect of each proposed modification.*

Response. Ablative experiments have been added to demonstrate the individual impact of each component of the solution.

Point 2. *Insufficient characterization of the mathematical implications of the cost function.*

Response. The mathematical section has been expanded to include detailed analysis of the cost function properties.

Point 3. *Lack of clarity in hyperparameter selection.*

Response. Optuna was used to select the hyperparameters through all experiments. The scalar experiment in particular was useful for determining the most meaningful hyperparameters for the model with the help of the Optuna B.

Point 4. *Improvements in the visual presentation of figures and tables.*

Response. All figures and tables have been reviewed and improved for greater clarity.

Point 5. *Consistency in mathematical notation.*

Response. All mathematical notation in the document has been reviewed and standardized.

Point 6. *Improvements in writing and text clarity.*

Response. The writing throughout the document has been reviewed and improved for greater clarity.

Conclusion

We have systematically addressed all the comments and recommendations provided by both reviewers. The document has been significantly improved in terms of:

- Structural cohesion and logical flow
- Bibliographic support and references
- Clarity in methodology and justification of decisions
- Presentation of results and analysis
- Visual quality of figures and tables
- Consistency in notation and writing

We thank you again for your valuable time and constructive feedback, which have been fundamental for elevating the quality of this research work.